

Access Remedies for Cross-Border Mergers in Network Industries

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Abstract

This paper develops a competition-economics model of access remedies for cross-border mergers in network industries. A merger between foreign firms may create network-scale efficiencies by integrating previously fragmented networks, but it may also reduce contestability in the domestic market. The paper compares three regimes: separate foreign firms, unconditional merger clearance, and merger clearance conditional on an access or interoperability remedy. The access remedy allows an independent entrant to compete with partial compatibility to the merged network. The model shows that such remedies can be welfare-superior to unconditional clearance when network effects are sufficiently strong, access quality is high, entrant costs are moderate, and implementation costs are limited. The mechanism is a scale-contestability trade-off: conditional clearance preserves part of the merged firm's network benefit while restoring competitive pressure through interoperable entry. The paper provides a formal rationale for access-based merger remedies in technology-intensive and platform-like markets where network advantages, compatibility, and entry viability are central competition concerns.

Keywords: Merger remedies; access remedies; interoperability; network effects; cross-border mergers; competition policy; contestability; Cournot oligopoly.

JEL Codes: L13; L41; L51; F13; D43.

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1 Introduction

Merger control in network industries requires authorities to evaluate more than the usual trade-off between concentration and cost efficiency. In these markets, the value of a product or service depends partly on the size and compatibility of the network. A merger may therefore generate genuine network-scale benefits by integrating technologies, users, or complementary systems. At the same time, the same integration may strengthen the merged firm's market position by increasing its network advantage and making entry more difficult. This creates a policy problem that is especially relevant in cross-border settings: a domestic authority may review, or be affected by, a foreign merger that changes domestic consumer surplus, domestic contestability, and the viability of local or independent entry.

A binary choice between prohibition and unconditional clearance may be too coarse in this environment. Blocking a merger may preserve the pre-merger market structure, but it may also sacrifice network-integration benefits. Unconditional clearance may preserve those benefits, but it can leave the domestic market dependent on a single integrated foreign firm. A third possibility is conditional clearance. The authority permits the merger but requires the merged firm to provide access, interoperability, or technology licensing on terms that allow an independent entrant to compete. This paper studies when such an access remedy is welfare-superior to unconditional merger clearance.

The central argument is that access remedies can separate two effects that are bundled together in an unconditional merger. A merger may increase the value of the network, but it may also reduce competitive pressure. An access remedy can preserve part of the network-scale benefit while restoring contestability through compatible entry. The relevant policy question is therefore not simply whether network effects make mergers more harmful. The question is whether the authority can design a remedy that converts part of the merged firm's network advantage into contestable market expansion.

The paper develops a simple model to formalize this mechanism. A Home market is initially served by two foreign firms. Consumers value the product more when the expected compatible network is larger. The paper compares three regimes. In the first regime, the foreign firms remain separate and compete in the Home market. In the second regime, the foreign firms merge and the merger is cleared unconditionally. In the third regime, the merger is cleared subject to an access or interoperability remedy that allows a domestic or independent entrant to compete with partial compatibility to the merged network.

Network expectations are treated in the standard fulfilled-expectations manner: firms choose quantities taking expected network size as given, and expectations are imposed after the quantity first-order conditions are solved. This is a technical modeling convention rather than the paper's contribution. The substantive contribution is to use this struc-

ture to study a merger-remedy problem: when does conditional clearance with access dominate unconditional clearance?

The model identifies a scale-contestability trade-off. Under unconditional clearance, the merged foreign firm integrates the network and may rationalize production through the more efficient technology. However, the merger also eliminates competition between the two foreign firms. Under conditional clearance, the access remedy allows an independent entrant to connect to the merged network with access quality ϕ . The entrant's participation restores contestability, while the compatibility parameter determines how much of the merged network's scale benefit is preserved.

The main result is that access remedies become more attractive as network effects strengthen, provided access is economically meaningful. In a clean analytical benchmark, the welfare gain from the access remedy relative to unconditional clearance is increasing in the network-effect parameter whenever access quality exceeds a weak lower bound. If implementation costs are positive but not too large, there exists a threshold n^* such that unconditional clearance is preferred for $n < n^*$, while conditional clearance with an access remedy is preferred for $n > n^*$. In the general model, the remedy must satisfy two conditions: it must increase Home welfare relative to unconditional clearance, and it must satisfy the merged firm's participation constraint.

The analysis yields three comparative-static predictions. First, higher access quality lowers the threshold at which the remedy becomes optimal because better interoperability makes entry more effective. Second, higher entrant cost raises the threshold because inefficient entry contributes less to contestability. Third, higher implementation or monitoring cost raises the threshold and may eliminate the remedy region. These results match the practical logic of competition remedies: access obligations are most attractive when access is technically meaningful, when a credible entrant exists, and when compliance can be monitored at reasonable cost.

The paper contributes to the competition-economics literature in three ways. First, it contributes to merger-remedy theory by studying access and interoperability remedies rather than standard structural divestitures. Second, it contributes to the analysis of digital and technology-intensive markets by showing how network effects can strengthen the case for conditional clearance rather than simply strengthening the case for prohibition. Third, it connects cross-border market-access concerns with merger-control practice by studying a domestic authority facing a foreign merger that affects domestic welfare and contestability.

The paper is organized as follows. Section 2 discusses the related literature. Section 3 presents the model and timing. Section 4 derives the equilibrium under the three regimes. Section 5 defines Home welfare and the remedy feasibility condition. Section 6 derives the network threshold for access-remedy dominance. Section 7 provides numerical illustrations. Section 8 discusses implications for competition-policy practice. Section 9

concludes.

2 Related Literature

This paper contributes to the competition-economics literature on merger remedies, network industries, and digital-market contestability. The closest literature is the economic analysis of merger remedies. Merger control often requires authorities to decide whether to prohibit a transaction, clear it unconditionally, or clear it subject to commitments. Motta et al. (2007) discuss the role of remedies in European Union merger practice and emphasize that commitments can preserve competition when outright prohibition is not the only available instrument. Cabral (2003) shows that, in markets with free entry, merger efficiencies may have ambiguous consumer-welfare effects because efficiencies can also affect post-merger entry incentives. Vasconcelos (2010) analyzes structural remedies in a Cournot framework where mergers are motivated by efficiency gains and may be approved subject to partial divestiture. Davies and Lyons (2007) provide an empirical and policy-oriented assessment of EU merger remedies. This paper differs from this literature by studying access and interoperability remedies rather than divestitures, and by focusing on markets where entry depends on compatibility with a network rather than only on production cost.

A second related literature studies network effects, compatibility, and standardization. Katz and Shapiro (1985) analyze competition in markets with network externalities and compatibility choices. Farrell and Saloner (1985) study standardization and compatibility when coordination affects market outcomes. Katz and Shapiro (1994) provide a broader discussion of systems competition and network effects. In the present paper, this literature supplies the standard modeling language for network effects and fulfilled expectations. The paper does not aim to extend the theory of fulfilled expectations itself. Instead, it applies a standard expectations convention to a merger-remedy problem: whether access to a compatible network can make conditional clearance welfare-superior to unconditional clearance.

The paper also speaks to recent competition-policy debates on digital markets. Digital and technology-intensive markets often involve network effects, data advantages, ecosystem linkages, switching costs, and economies of scale and scope. These features can make entry difficult and can increase the importance of interoperability and access. Crémer et al. (2019) argue that competition policy in the digital era must account for the role of data, platforms, and market power in dynamic environments. The Furman Report (Furman et al., 2019) emphasizes contestability and recommends pro-competitive interventions such as data mobility and open standards. OECD (2021) discuss how data portability and interoperability can reduce lock-in, facilitate multi-homing, and promote competition, while also noting that such measures may not be effective in every market.

This paper provides a formal model of one part of that policy discussion: access remedies are valuable when they restore contestability without destroying network-scale benefits.

The paper is also motivated by merger-control practice in digital and technology markets. Recent competition enforcement has paid increasing attention to foreclosure, ecosystem effects, data access, interoperability, and the possibility that a merged firm may weaken rivals by restricting access to important inputs or interfaces. The 2023 DOJ/FTC Merger Guidelines include a specific discussion of platform mergers and recognize that network effects and scale economies can be barriers to entry (U.S. DOJ and FTC, 2023). The European Commission’s digital-merger practice similarly highlights the challenges raised by vertical and conglomerate effects in digital and technology-focused sectors (European Commission, 2022). Cases such as Microsoft/LinkedIn and Google/Fitbit illustrate that digital merger review can involve behavioral or access-related commitments designed to preserve rival opportunities, data access, or interoperability (European Commission, 2016, 2020). The present model abstracts from case-specific institutional details but captures the common economic issue: a merger may create network benefits while simultaneously weakening the conditions for entry.

Finally, the paper has a secondary connection to cross-border competition policy and strategic trade. A domestic authority may care about domestic consumer surplus, domestic entry, and market contestability even when the merging firms are foreign. This creates an overlap between merger control and market-access concerns. However, the paper is not primarily a strategic-trade model. Tariffs and cross-border ownership provide the institutional environment, while the substantive contribution is to merger-remedy theory. The main question is whether conditional clearance through access or interoperability can dominate unconditional clearance in a network industry.

The contribution of the paper is therefore to connect three strands of work: the economics of merger remedies, the economics of network compatibility, and competition-policy practice in digital and technology-intensive markets. It shows that network effects do not necessarily imply that mergers should be blocked. When access quality is high, entrant costs are moderate, and remedy implementation costs are limited, stronger network effects can increase the case for conditional clearance because access remedies preserve network-scale benefits while restoring contestability.

3 The Model

Consider a Home market served initially by two foreign firms, indexed by $i = 1, 2$. The Home country has consumers but no domestic producer at the outset. Firm 1 is the efficient foreign firm and firm 2 is the less efficient foreign firm, with marginal costs

$$c_1 < c_2. \tag{1}$$

The industry is characterized by network effects. Consumers' willingness to pay depends not only on current output but also on the expected size of the compatible network. Let Q denote total output supplied to the Home market and let Z^e denote the expected compatible network size. The inverse demand function is

$$P = a - Q + nZ^e, \quad (2)$$

where $a > 0$ is market size and $n \in (0, 1)$ measures the strength of the network effect.

Firms take the expected network size Z^e as given when choosing quantities. Fulfilled expectations are imposed only after the firms' first-order conditions have been derived and solved. Thus, we do not substitute $Z^e = Z(q)$ directly into demand before taking first-order conditions.

The paper compares three regimes. In Regime I, the foreign firms remain separate and compete in the Home market. In Regime II, the foreign firms merge and the merger is cleared unconditionally. In Regime III, the foreign merger is cleared subject to an access or interoperability remedy that enables a domestic or independent entrant to compete using partial compatibility with the merged network.

3.1 Timing

The game has three stages.

- (i) The market structure is one of three regimes: separate foreign firms, unconditional foreign merger, or foreign merger with an access remedy.
- (ii) Conditional on the regime, the Home authority chooses the relevant tariff and, under the remedy regime, the access parameters. In the baseline analysis, these instruments are treated as policy parameters.
- (iii) Active firms choose quantities in the Home market. Firms take expected compatible network size as fixed. After quantity first-order conditions are solved, fulfilled expectations are imposed.

Assumption 1 (Interior equilibrium). *The parameter values are such that all quantities considered in the main analysis are strictly positive. Corner cases are discussed by implication through the entrant-viability condition.*

Assumption 2 (Fulfilled-expectations discipline). *In every regime, firms choose quantities taking Z^e as fixed. Fulfilled expectations are imposed only after the quantity first-order conditions have been solved.*

4 Fulfilled-Expectations Equilibrium

4.1 A Useful Cournot Lemma

Consider two active firms i and k , with marginal costs m_i and m_k . Let their quantities be q_i and q_k . Total output is $Q = q_i + q_k$, and inverse demand is $P = a - q_i - q_k + nZ^e$. Suppose the fulfilled compatible network size is

$$Z = \omega_i q_i + \omega_k q_k, \quad (3)$$

where $\omega_i \geq 0$ and $\omega_k \geq 0$ measure the contribution of each firm's output to the compatible network.

Since Z^e is fixed at the quantity-choice stage, the first-order conditions are

$$a - 2q_i - q_k + nZ^e - m_i = 0, \quad (4)$$

$$a - q_i - 2q_k + nZ^e - m_k = 0. \quad (5)$$

Solving gives

$$q_i(Z^e) = \frac{a + nZ^e - 2m_i + m_k}{3}, \quad (6)$$

$$q_k(Z^e) = \frac{a + nZ^e - 2m_k + m_i}{3}. \quad (7)$$

Fulfilled expectations require $Z^e = \omega_i q_i(Z^e) + \omega_k q_k(Z^e)$. Hence

$$Z^* = \frac{(\omega_i + \omega_k)a + (-2\omega_i + \omega_k)m_i + (\omega_i - 2\omega_k)m_k}{3 - n(\omega_i + \omega_k)}. \quad (8)$$

The equilibrium quantities are then obtained from (6) and (7) using Z^* .

Lemma 1 (Two-firm fulfilled-expectations Cournot equilibrium). *Suppose $3 - n(\omega_i + \omega_k) > 0$. Then the two-firm network Cournot game has a unique fulfilled-expectations equilibrium given by (8) and (6)-(7).*

Proof. The firms' first-order conditions form a linear system in quantities for a fixed Z^e . Substituting the quantity functions into the fulfilled-expectations condition gives a single linear equation in Z^e . The denominator is positive by assumption, so the solution is unique. $\square$

4.2 Regime I: Separate Foreign Firms

In Regime I, the foreign firms remain separate. The Home authority may impose firm-specific tariffs τ_1 and τ_2 . Effective marginal costs are

$$m_1 = c_1 + \tau_1, \quad (9)$$

$$m_2 = c_2 + \tau_2. \quad (10)$$

The two foreign networks are not fully integrated. Let $\lambda \in (0, 1)$ measure the degree of compatibility between the two separate foreign networks. The expected compatible network is

$$Z_I^e = \lambda(q_1^e + q_2^e). \quad (11)$$

Profits are

$$\pi_1^I = (a - q_1 - q_2 + nZ_I^e - m_1)q_1, \quad (12)$$

$$\pi_2^I = (a - q_1 - q_2 + nZ_I^e - m_2)q_2. \quad (13)$$

Using Lemma 1 with $\omega_1 = \omega_2 = \lambda$, the fulfilled-expectations network size is

$$Z_I^* = \frac{\lambda(2a - m_1 - m_2)}{3 - 2n\lambda}. \quad (14)$$

Equilibrium quantities are

$$q_1^{I*} = \frac{a + nZ_I^* - 2m_1 + m_2}{3}, \quad (15)$$

$$q_2^{I*} = \frac{a + nZ_I^* - 2m_2 + m_1}{3}. \quad (16)$$

Total output is

$$Q_I^* = q_1^{I*} + q_2^{I*}. \quad (17)$$

Proposition 1 (Separate foreign-firm equilibrium). *In Regime I, if $3 - 2n\lambda > 0$, there exists a unique fulfilled-expectations Cournot equilibrium. The equilibrium compatible network is given by (14), and equilibrium quantities are given by (15)-(16).*

4.3 Regime II: Foreign Merger with Unconditional Clearance

In Regime II, the two foreign firms merge. The merged firm is denoted by M . The merger allows production to be rationalized through the efficient technology, so the merged firm's marginal cost is c_1 . The Home authority applies a tariff τ_M to the merged foreign firm. Thus,

$$m_M = c_1 + \tau_M. \quad (18)$$

The merger fully integrates the foreign network, so

$$Z_{II}^e = q_M^e. \quad (19)$$

The merged firm's profit is

$$\pi_M^{II} = (a - q_M + nZ_{II}^e - m_M)q_M. \quad (20)$$

Taking Z_{II}^e as fixed, the first-order condition is

$$a - 2q_M + nZ_{II}^e - m_M = 0. \quad (21)$$

Hence

$$q_M^{II}(Z_{II}^e) = \frac{a + nZ_{II}^e - m_M}{2}. \quad (22)$$

Fulfilled expectations require $Z_{II}^e = q_M^{II}$. Therefore,

$$q_M^{II*} = \frac{a - m_M}{2 - n}. \quad (23)$$

Since the merged firm is the only active producer, $Q_{II}^* = q_M^{II*}$.

Proposition 2 (Unconditional merger equilibrium). *In Regime II, the unique fulfilled-expectations Cournot equilibrium is*

$$q_M^{II*} = \frac{a - m_M}{2 - n}. \quad (24)$$

Remark 1. *The denominator is $2 - n$, not $2(1 - n)$. The latter would arise only if one incorrectly substituted fulfilled expectations into demand before deriving the firm's first-order condition.*

4.4 Regime III: Foreign Merger with an Access Remedy

In Regime III, the foreign merger is cleared only if the merged firm grants access or interoperability to a domestic or independent entrant, denoted by D . The merged firm and the entrant then compete in quantities.

The merged firm's effective marginal cost is

$$m_M^R = c_1 + \tau_R, \quad (25)$$

where τ_R is the tariff imposed on the foreign merged firm under the remedy regime. The domestic entrant obtains access at a per-unit access charge ρ . Its effective marginal cost

is

$$m_D = c_D + \rho. \quad (26)$$

The quality of the access remedy is measured by $\phi \in (0, 1]$. If $\phi = 1$, the entrant receives full compatibility with the merged network. If $\phi < 1$, access is imperfect. The expected compatible network is

$$Z_R^e = q_M^e + \phi q_D^e. \quad (27)$$

Profits are

$$\pi_M^R = (a - q_M - q_D + nZ_R^e - m_M^R)q_M + \rho q_D, \quad (28)$$

$$\pi_D^R = (a - q_M - q_D + nZ_R^e - m_D)q_D. \quad (29)$$

At the quantity stage, Z_R^e , ρ , ϕ , and τ_R are fixed. The access revenue term ρq_D does not affect the merged firm's first-order condition with respect to q_M , because q_D is chosen by the entrant. The first-order conditions are

$$a - 2q_M - q_D + nZ_R^e - m_M^R = 0, \quad (30)$$

$$a - q_M - 2q_D + nZ_R^e - m_D = 0. \quad (31)$$

Using Lemma 1 with $\omega_M = 1$ and $\omega_D = \phi$, the fulfilled-expectations network size is

$$Z_R^* = \frac{(1 + \phi)a + (-2 + \phi)m_M^R + (1 - 2\phi)m_D}{3 - n(1 + \phi)}. \quad (32)$$

Equilibrium quantities are

$$q_M^{R*} = \frac{a + nZ_R^* - 2m_M^R + m_D}{3}, \quad (33)$$

$$q_D^{R*} = \frac{a + nZ_R^* - 2m_D + m_M^R}{3}. \quad (34)$$

Equivalently, defining $D_R(n, \phi) = 3 - n(1 + \phi)$,

$$q_M^{R*} = \frac{a + m_D - 2m_M^R + n\phi(m_M^R - m_D)}{D_R(n, \phi)}, \quad (35)$$

$$q_D^{R*} = \frac{a + m_M^R - 2m_D + n(m_D - m_M^R)}{D_R(n, \phi)}, \quad (36)$$

$$Q_R^* = \frac{2a - m_D - m_M^R + n(1 - \phi)(m_D - m_M^R)}{D_R(n, \phi)}. \quad (37)$$

Proposition 3 (Access-remedy equilibrium). *In Regime III, if $3 - n(1 + \phi) > 0$, there exists a unique fulfilled-expectations Cournot equilibrium. The equilibrium compatible network is given by (32), and equilibrium quantities are given by (33)-(34).*

5 Home Welfare and Remedy Feasibility

Home welfare includes consumer surplus, tariff revenue, and the profit of the domestic entrant when such an entrant exists. Foreign profits are excluded from Home welfare. Since inverse demand is linear with slope -1 , consumer surplus in any regime j is

$$CS_j = \frac{1}{2}(Q_j^*)^2. \quad (38)$$

In Regime I,

$$SW_I = \frac{1}{2}(Q_I^*)^2 + \tau_1 q_1^{I*} + \tau_2 q_2^{I*}. \quad (39)$$

In Regime II,

$$SW_{II} = \frac{1}{2}(q_M^{II*})^2 + \tau_M q_M^{II*}. \quad (40)$$

In Regime III,

$$SW_R = \frac{1}{2}(Q_R^*)^2 + \tau_R q_M^{R*} + \pi_D^{R*} - \Psi(\phi), \quad (41)$$

where $\Psi(\phi)$ is the implementation or monitoring cost of the access remedy. From the entrant's first-order condition, equilibrium entrant profit is

$$\pi_D^{R*} = (q_D^{R*})^2. \quad (42)$$

Thus,

$$SW_R = \frac{1}{2}(Q_R^*)^2 + \tau_R q_M^{R*} + (q_D^{R*})^2 - \Psi(\phi). \quad (43)$$

An access remedy is not automatically feasible. The merged foreign firm must be willing to accept conditional clearance rather than take its outside option. The merged firm's profit under the remedy is

$$\pi_M^{R*} = (q_M^{R*})^2 + \rho q_D^{R*}, \quad (44)$$

where the equality uses the merged firm's first-order condition. Let Π^{out} denote the merged firm's relevant outside option. The participation constraint is

$$(q_M^{R*})^2 + \rho q_D^{R*} \geq \Pi^{out}. \quad (45)$$

Proposition 4 (Desirability and feasibility). *Conditional clearance with an access remedy is preferred to unconditional merger clearance only if it is both welfare-desirable and feasible. Welfare desirability requires $SW_R \geq SW_{II}$, while feasibility requires (45).*

6 Welfare Comparison and Threshold Result

This section compares unconditional merger clearance with conditional clearance through an access remedy. The central welfare difference is

$$\Delta_R(n) = SW_R(n) - SW_{II}(n). \quad (46)$$

6.1 A Clean Benchmark

To obtain a transparent analytical result, consider a benchmark in which the merged firm and the entrant have the same effective marginal cost under the remedy:

$$m_M = m_M^R = m_D = m. \quad (47)$$

Let $A = a - m > 0$. Under unconditional clearance,

$$q_M^{II*} = \frac{A}{2 - n}. \quad (48)$$

Under the access remedy,

$$q_M^{R*} = q_D^{R*} = \frac{A}{3 - n(1 + \phi)}, \quad (49)$$

and total output is

$$Q_R^* = \frac{2A}{3 - n(1 + \phi)}. \quad (50)$$

Let the implementation cost of the remedy be fixed at $K \geq 0$. With tariffs suppressed in this benchmark,

$$SW_{II} = \frac{1}{2} \left(\frac{A}{2 - n} \right)^2, \quad (51)$$

and

$$SW_R = \frac{3A^2}{[3 - n(1 + \phi)]^2} - K. \quad (52)$$

Therefore,

$$\Delta_R(n) = A^2 \left[\frac{3}{[3 - n(1 + \phi)]^2} - \frac{1}{2(2 - n)^2} \right] - K. \quad (53)$$

6.2 Access Quality and Monotonicity

Differentiating (53) with respect to n gives

$$\frac{\partial \Delta_R(n)}{\partial n} = A^2 \left[\frac{6(1 + \phi)}{[3 - n(1 + \phi)]^3} - \frac{1}{(2 - n)^3} \right]. \quad (54)$$

Define ϕ_0 as the unique root in $(0, 1)$ of

$$\phi^3 - 6\phi^2 + 18\phi - 2 = 0. \quad (55)$$

Numerically,

$$\phi_0 \simeq 0.11547. \quad (56)$$

Lemma 2 (Sufficient access quality). *If $\phi \geq \phi_0$, then*

$$\frac{\partial \Delta_R(n)}{\partial n} > 0 \quad \text{for all } n \in (0, 1). \quad (57)$$

Proof. From (54), $\partial \Delta_R(n)/\partial n > 0$ if and only if

$$G(n, \phi) = 6(1 + \phi)(2 - n)^3 - [3 - n(1 + \phi)]^3 > 0. \quad (58)$$

For $n \in (0, 1)$ and $\phi \in (0, 1]$, the minimum of $G(n, \phi)$ over n is attained at $n \rightarrow 1$. At $n = 1$,

$$G(1, \phi) = 6(1 + \phi) - (2 - \phi)^3 = \phi^3 - 6\phi^2 + 18\phi - 2. \quad (59)$$

Therefore, $G(1, \phi) \geq 0$ if and only if $\phi \geq \phi_0$. Hence $G(n, \phi) > 0$ for all $n \in (0, 1)$ when $\phi \geq \phi_0$. $\square$

Proposition 5 (Network threshold for access-remedy dominance). *Suppose the clean benchmark holds and access quality satisfies $\phi \geq \phi_0$. If the implementation cost K satisfies*

$$\frac{5A^2}{24} < K < A^2 \left[\frac{3}{(2 - \phi)^2} - \frac{1}{2} \right], \quad (60)$$

then there exists a unique threshold $n^ \in (0, 1)$ such that*

$$SW_R(n) > SW_{II}(n) \quad \text{if and only if } n > n^*. \quad (61)$$

Proof. From (53),

$$\Delta_R(0) = \frac{5A^2}{24} - K. \quad (62)$$

Thus $\Delta_R(0) < 0$ when $K > 5A^2/24$. At $n = 1$,

$$\Delta_R(1) = A^2 \left[\frac{3}{(2 - \phi)^2} - \frac{1}{2} \right] - K. \quad (63)$$

Thus $\Delta_R(1) > 0$ when the upper bound in (60) holds. The lemma establishes that $\Delta_R(n)$ is strictly increasing in n when $\phi \geq \phi_0$. Continuity and strict monotonicity imply a unique $n^* \in (0, 1)$. $\square$

Remark 2 (Interpretation of the access-quality condition). *The condition $\phi \geq \phi_0$ is*

weak. Since $\phi_0 \simeq 0.11547$, the condition only rules out extremely poor access remedies. Economically, this is desirable. If the access remedy gives the entrant almost no meaningful compatibility with the merged network, stronger network effects need not make the remedy more valuable. The theorem therefore formalizes a practical point: access remedies work only when access is economically meaningful.

7 Numerical Illustration

The analytical results show that access remedies can dominate unconditional merger clearance when network effects are sufficiently strong. This section provides a numerical illustration. The purpose is not calibration. Rather, it demonstrates the comparative statics implied by the model: stronger network effects raise the value of access remedies, higher access quality lowers the threshold at which the remedy becomes optimal, and higher entrant costs weaken the case for the remedy.

The baseline parameter values are

$$a = 10, \quad m_M = m_M^R = 2, \quad m_D = 2, \quad \phi = 0.8, \quad K = 30, \quad (64)$$

and tariffs are set to zero. Under these values, the threshold is approximately $n^* \simeq 0.512$. Thus, unconditional clearance is preferred at low network intensity, while the access remedy dominates once network effects become sufficiently strong.

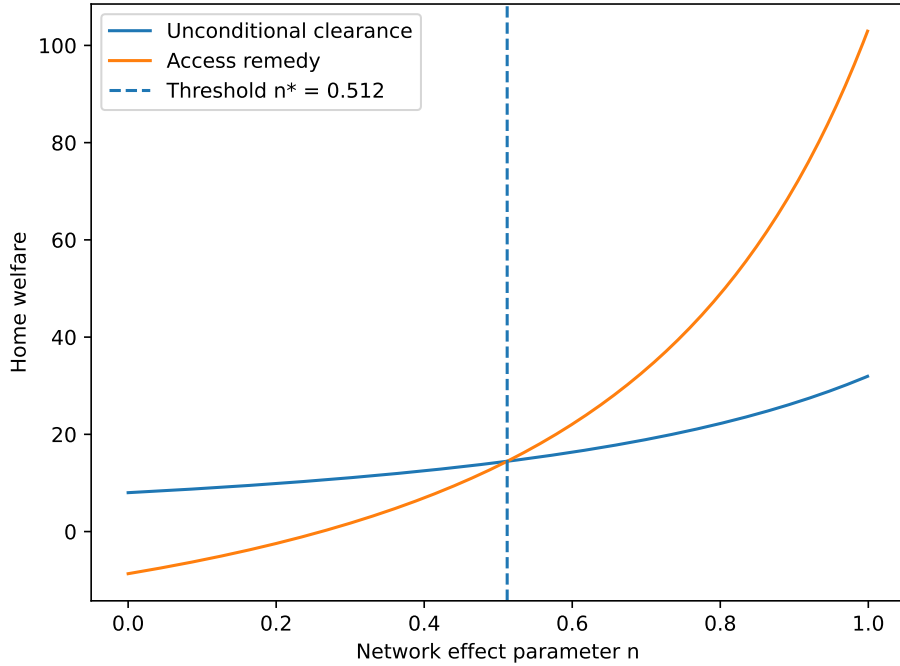


Figure 1: Welfare comparison across network intensity.

Figure 1 shows the welfare comparison between unconditional clearance and the access

remedy. The access remedy is initially inferior because the implementation cost is not justified by weak network effects. As n rises, interoperable entry becomes more valuable and the remedy eventually dominates.

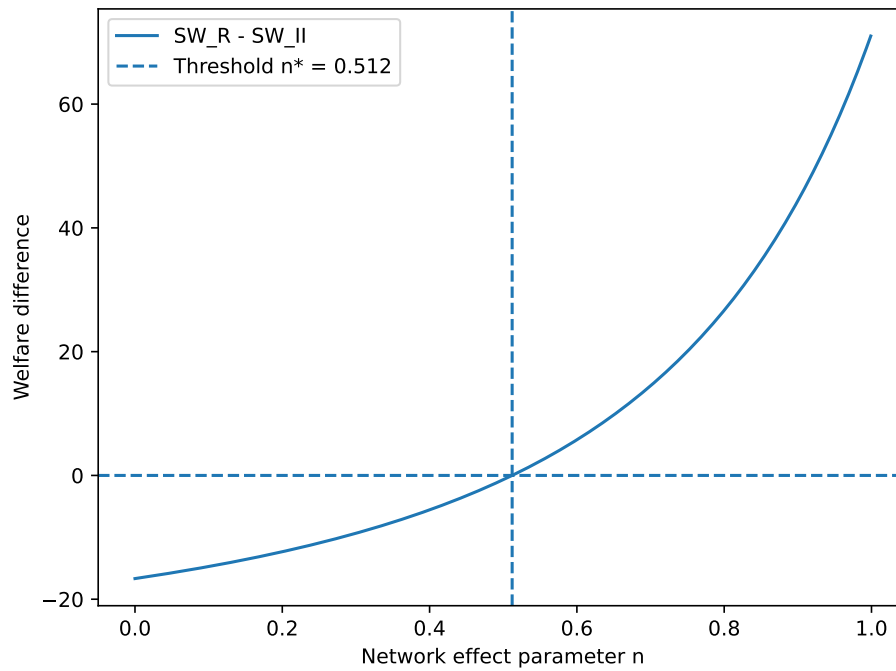


Figure 2: Welfare difference $SW_R - SW_{II}$.

Figure 2 plots the welfare difference directly. The zero crossing identifies the threshold n^* .

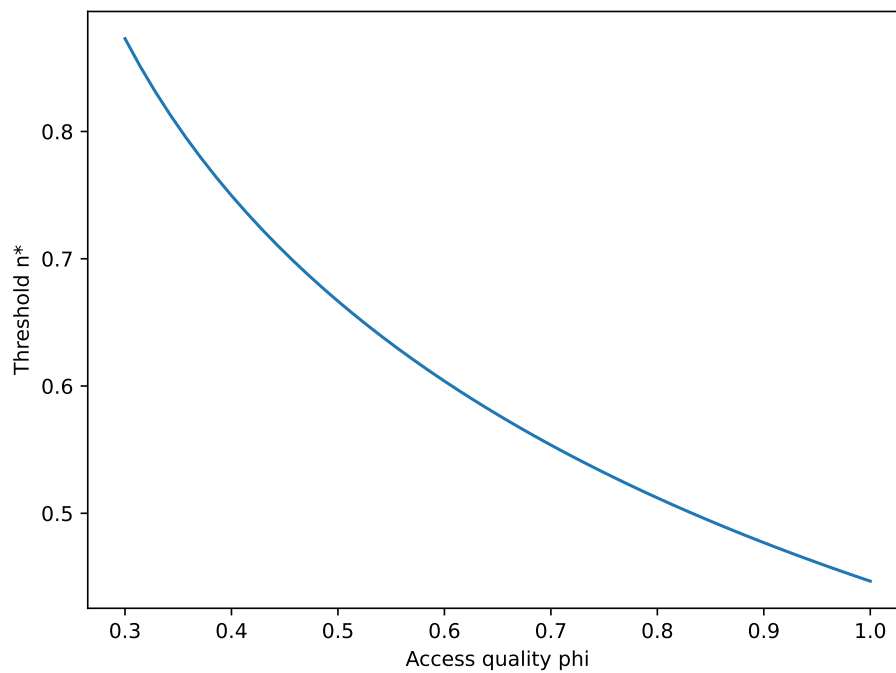


Figure 3: Access quality and the remedy threshold.

Figure 3 varies access quality ϕ . Higher access quality lowers the network threshold. Economically, better interoperability makes entry more effective and allows the entrant to contribute more to the compatible network.

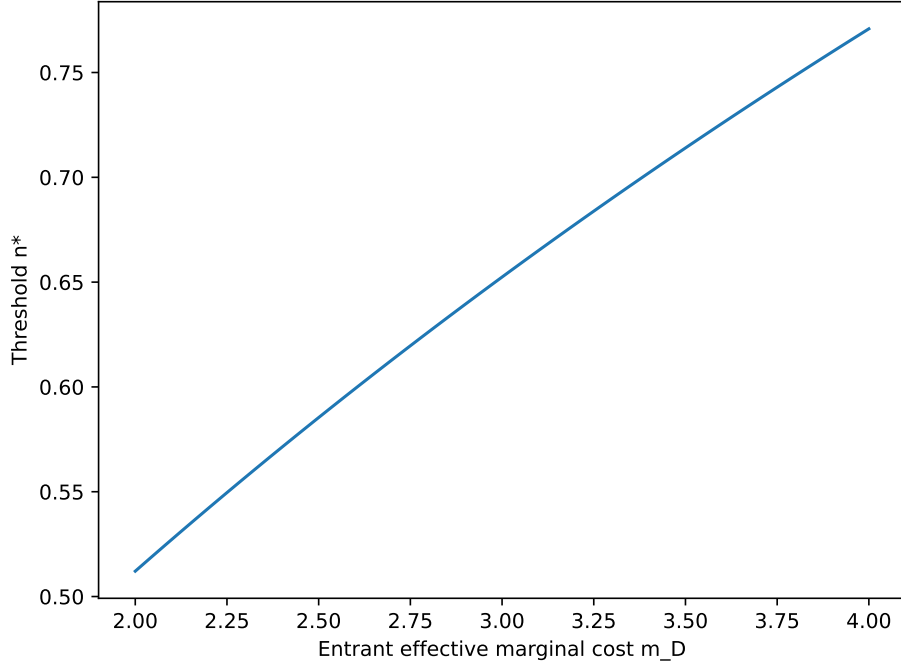


Figure 4: Entrant cost and the remedy threshold.

Figure 4 varies the entrant's effective marginal cost m_D . Higher entrant cost raises the threshold because entry contributes less to contestability. If entrant cost is sufficiently high, the access remedy may not dominate over the relevant range of network effects.

8 Policy Discussion

The model provides a competition-economics rationale for access remedies in network industries. Its main implication is not that network effects always make mergers harmful, nor that authorities should routinely impose access obligations. Rather, the analysis identifies conditions under which conditional clearance can dominate unconditional clearance. Access remedies are most attractive when they preserve merger-specific network benefits while restoring contestability through viable entry.

8.1 Conditional Clearance versus Unconditional Clearance

In standard merger analysis, unconditional clearance may be appropriate when efficiency gains are substantial and the loss of competition is limited. In a network industry, however, the same merger may have two opposing effects. It may increase value by

integrating networks, but it may also make entry more difficult by strengthening the merged firm's installed-base advantage. The model captures this as a scale-contestability trade-off.

Unconditional clearance preserves network integration but leaves the merged firm as the sole beneficiary of that integration. Conditional clearance changes the allocation of the network benefit. By requiring access or interoperability, the authority allows an entrant to share part of the compatible network. The remedy therefore converts some of the merged firm's network advantage into a source of contestable entry.

8.2 Access Quality and Entrant Viability

A central policy variable in the model is access quality, denoted by ϕ . This parameter captures the extent to which the entrant can benefit from the merged firm's network. In practice, access quality may depend on the technical and contractual design of the remedy: the scope of interoperability, the quality of APIs, latency, data format, non-discrimination rules, update frequency, dispute-resolution procedures, and protection against strategic degradation.

The model predicts that higher access quality lowers the network threshold at which the remedy becomes welfare-superior to unconditional clearance. The policy meaning is straightforward. A weak or symbolic access remedy may not restore contestability, especially if the entrant cannot provide users with a service that is meaningfully compatible with the merged network. By contrast, a high-quality access remedy makes entry more effective and allows the authority to preserve network benefits while reducing post-merger market power.

The model also shows that access remedies require a credible entrant. The entrant's effective marginal cost is $m_D = c_D + \rho$, where c_D is the entrant's production cost and ρ is the access charge. Access obligations cannot create competition by themselves if no entrant has the capability, scale, or commercial incentive to use the access effectively.

8.3 Access Charges and Feasibility

The access charge ρ plays two roles. On the one hand, a higher access charge raises the entrant's effective marginal cost and weakens entry. On the other hand, it compensates the merged firm and can help satisfy the participation constraint. The merged firm accepts the remedy only if

$$(q_M^{R*})^2 + \rho q_D^{R*} \geq \Pi^{out}. \quad (65)$$

This captures an important feature of practical merger remedies. Remedies must be welfare-desirable, but they must also be implementable. If the access price is too low, the remedy may be attractive for the entrant but unacceptable to the merged firm. If

the access price is too high, the remedy may satisfy the merged firm but fail to generate entry.

8.4 Implementation and Monitoring Costs

The model includes a remedy implementation cost $\Psi(\phi)$. This cost represents technical implementation, compliance monitoring, interface maintenance, dispute resolution, and authority oversight. Access remedies are not costless. Higher implementation costs make access remedies less attractive. If implementation costs are too high, unconditional clearance or prohibition may be preferable. This is an important limit on the paper's policy conclusion. The model does not imply that access remedies should be imposed whenever network effects are present. It implies that they are most attractive when the authority can define, monitor, and enforce the remedy at reasonable cost.

8.5 Implications for Cross-Border Merger Review

The cross-border setting matters because the merging firms may be foreign while the welfare effects occur in the domestic market. A Home authority may care about domestic consumer surplus, domestic contestability, and the viability of local or independent entry, even if the merging firms' profits are foreign. In such cases, unconditional clearance may transfer network advantages to a foreign merged firm without preserving domestic competitive constraints.

An access remedy can address this concern without blocking the transaction outright. It permits the merger but conditions the merger's domestic effect on access terms that preserve contestability in the Home market. This does not require the authority to treat foreign ownership as harmful per se. The competition concern is not nationality. The concern is whether the merger creates or strengthens a network position that makes domestic entry less viable.

The broader implication is that network effects should not lead mechanically to either prohibition or unconditional clearance. In some markets, strong network effects strengthen the case for a well-designed conditional clearance remedy. The authority's task is to determine whether access can transform the merged network from a barrier to entry into a platform for contestable competition.

9 Conclusion

This paper develops a competition-economics model of access remedies for cross-border mergers in network industries. The starting point is a practical merger-control problem. In markets with network effects, a merger may create real efficiency gains by integrating

previously fragmented networks, but it may also weaken contestability by strengthening the merged firm’s network advantage. A binary policy choice between prohibition and unconditional clearance may therefore be too coarse. Conditional clearance with an access or interoperability remedy can sometimes preserve network-scale benefits while restoring competitive entry.

The model compares three regimes: separate foreign firms, unconditional merger clearance, and merger clearance conditional on an access remedy. The remedy allows an independent entrant to compete with partial compatibility to the merged network. The analysis identifies a scale-contestability trade-off. Unconditional clearance preserves the merged firm’s integrated network but does not restore competition. Conditional clearance reduces the merged firm’s exclusive control over the network advantage and allows part of that advantage to support contestable entry.

The main result is that access remedies become more attractive as network effects strengthen, provided that access quality is sufficiently high, entrant costs are moderate, and implementation costs are limited. In the clean analytical benchmark, there exists a threshold level of network effects above which conditional clearance dominates unconditional clearance. In the general model, the access remedy must satisfy two conditions. It must be welfare-desirable for the Home country, and it must be feasible for the merged firm, as captured by the participation constraint. This distinction is important because merger remedies are not costless commands. They must be designed so that they both improve welfare and can be implemented in practice.

The paper is intentionally simple. It abstracts from dynamic investment, multi-sided pricing, bargaining over access terms, data accumulation, and strategic degradation of interoperability. These are natural extensions. A dynamic version could study how access remedies affect innovation incentives and installed-base accumulation over time. A platform version could allow different user groups and cross-side network effects. A regulatory-design extension could endogenize the access price and study FRAND-type obligations more explicitly. These extensions would enrich the framework, but the baseline model already delivers a clear message: in network industries, well-designed access remedies can transform a merger-generated network advantage from a barrier to entry into a basis for contestable competition.

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A Replication Code

The Python code used to generate the numerical illustrations is included in the accompanying file `code/generate_figures.py`.